# GUIDELINES FOR PREPERATION OF MCA PROJECT REPORT

## Preamble

While utmost attention must be paid to the content of the Project report, which is being submitted in partial fulfilment of the requirements of the MCA degree, it is imperative that a standard format be prescribed. The same format shall also be followed in preparation of the final soft copies to be submitted to the Library.

| Serial No | Topics to be Covered | Page Number |
|---|---|---|
| 1 | **Introduction of the Project** | |
| 2 | **Concepts and Problem Analysis**<br><br>Cost Estimation (cocomo model / Function Point)<br>Time Analysis<br>Team structure<br>Software Configuration Management<br>Quality Assurance Plan<br>Risk Management | |
| 3 | **Web application Survey and use of Inspect tool** | |
| 4 | **Theoritical Background** | |
| 5 | **Software and Hardware Requirements, Client and Development Environment Specifications, and Error Handling Guidelines** | |
| 6 | **Design Methodology**<br>Data Design<br>Architectural Design ( Preferable UML Design)<br>Interface Design<br>Procedural design | |
| 7 | **Coding standard and Assumptions Folloowed** | |
| 8 | **Testing and Test Cases** | |
| 9 | **Conclusion** | |
| 10 | **Future Scope of the Project** | |
| 11 | **References and Bibiliography** | |

**Note:** Students are advised that the complete code of the project should **not** be included within the main body of the project report. Instead, a detailed code listing and the project hierarchy for a specific module should be provided in the Appendices.

When presenting the *Procedural Design*, you are expected to describe each class in the module comprehensively. This should include the class name, its purpose, and a summary of the methods contained within the class. For each method or function, you must specify its purpose, the inputs it requires, and the outputs it produces.

Proper annotation of the code is strongly recommended. Where annotations are included, you may make use of appropriate tools to generate procedural documentation, thereby enhancing clarity and readability. This approach ensures that the report remains concise while providing sufficient technical detail for evaluation and replication of the work.

# 1. PROJECT REPORT FORMAT

## 1.1 Paper

**1.1.1 Quality:** The project report shall be printed / photo copied on white bond paper, whiteness 95% or above, weight 70 gram or more per square meter.

**1.1.2 Size:** The size of the paper shall be standard A4; height 297 mm, width 210mm.

**1.1.3 Type Setting, Text Processing and Printing:** The text shall be printed on single sides employing LaserJet or Inkjet printer, the text having been processed using a standard text processor. The standard font shall be Times New Roman of 12 pts with 1.5-linespacing.

**1.1.1 Page Format:** Printed Sheets shall have the following written area and margins:
Top Margin 1''
Bottom Margin 1''
Left Margin 1''
Right Margin 1''
Single sided

**1.1.2 Pagination:** Page numbering in the text of the report shall be Hindu Arabic numerals at the center of the footer. But when the candidate opts for header style the page number shall appear at the right and left top corner for the odd and even number pages, respectively. Page number "1" for the first page of the Introduction chapter shall not appear in print, only the second page will bear the number "2". The subsequent chapters shall begin on a fresh page. When header style is chosen the first page of each chapter will not have the header and the page number shall be printed at the center of the footer.

**1.1.3 Header:** When the header style is chosen, the header can have the Chapter number and Section number (e.g., Chapter 2, Section 3) on even numbered page headers and Chapter title or Section title on the odd numbered page header.

**1.1.4 Paragraph format:** Vertical space between paragraphs shall be about 2.5-line spacing. The first line of each paragraph should normally be indented by five characters or 12mm. A candidate may, however, choose not to indent if he/she has provided sufficient paragraph separation. A paragraph should normally comprise more than one line. A single line of a paragraph shall not be left at the top or bottom of a page (that is, no windows or orphans should be left). The word at the right end of the first line of a page or paragraph should, as far as possible, not be hyphenated.

## 1.2 Chapter and Section Format

**1.2.1 Chapter:** Each chapter shall begin on a fresh page with an additional top margin of about 75mm. Chapter number (in Hindu-Arabic) and title shall be printed at the center of the line in 6mm font size (18pt) in bold face using both upper and lower case (all capitals or small capitals shall not be used). A vertical gap of about 25mm shall be left between the Chapter number and Chapter title lines and between chapter title line and the first paragraph.

**Sections and Subsections:** A chapter may be divided into *Sections*, *Subsections*, and *Sub-subsections* in order to present different concepts distinctly and in a well-structured manner. Sections and Subsections shall be numbered using decimal notation — for example, *2.2* for the second section in Chapter 2, and *2.3.4* for the fourth subsection under the third section of Chapter 2.
All *Chapters*, *Sections*, and *Subsections* shall be listed in the *Table of Contents*, with corresponding page numbers aligned to the right margin. Further subdivisions beyond the subsection level need not be numbered, nor should they appear in the contents.
Section and Subsection titles, together with their respective numbers, shall be presented in bold typeface and aligned to the left margin (not centred). The font size shall be 16 pt (5 mm) for *Section* titles and 14 pt (4 mm) for *Subsection* titles, with a spacing of 15 mm above and below each title line.
In the case of further subdivisions, a smaller font size—ranging between 3 mm and 3.5 mm—may be employed. Such titles may be formatted in bold, small capitals, full capitals, or italics, and may be either left-aligned or centred as appropriate. These lower-level subdivisions shall not be included in the *Table of Contents*.

**1.2.2 Table / Figure Format:** As far as possible, tables and figures should be presented in portrait style. Small size table and figures (less than half of writing area of a page) should be incorporated within the text, while larger ones may be presented on separate pages. Table and figures shall be numbered chapter-wise.

## 3 AUXILIARY FORMAT

**3.1 Binding:** The evaluation copies of the project report may be spiral bound or soft or hard bound. The final hard bound copies need to be submitted after the viva-voce examination. with the following color specification:

**3.2 Front Covers:** The front covers shall contain the following details:
Full title of report in 6 mm 22 point's size font properly centered and positioned at the top. Full name of the candidate(s) in 4.5 mm 15 point's size font properly centered at the middle of the page. A 40 mm diameter replica of the Institute emblem followed by the name of department, name of the Institute and the year of submission, each in a separate line and properly centered and located at the bottom of page.

**3.3 Blank Sheets:** In addition to the white sheets (binding requirement) two white sheets shall be put at the beginning and the end of the report.

**3.4 Title Sheet:** This shall be the first printed page of the Project report and shall contain the submission statement: The Project Report submitted in partial fulfilment of the requirements of the MCA Degree, the name and Roll No. of the candidate, name(s) of the Supervisor and Co-supervisor(s) (if any), Department, Institute and year of submission.
Sample copy of the 'Title Sheet' is appended (Specimen 'A').

**Approval Sheet:** The Recommendation Page shall constitute the first page of the project report. In this case, it shall not carry a page number. If it does not form the first page, it shall be assigned page number two, indicated in lowercase Roman numerals (“ii”) and positioned at the center of the footer. The top line of this page shall read: “Recommendation for MCA Project Report”. Students are advised to refer to the appended sample copies of the Recommendation and Approval Sheet for guidance, designated as Specimen ‘D’ and Specimen ‘B’. The page should be formatted clearly and professionally, following the provided specimen.

**3.5 Acknowledgements:** The acknowledgments by the candidate

**3.6 Abstract:** The 500-word abstract shall highlight the important features of the Project report

**3.7 Contents:** The contents shall follow the Abstract and shall enlist the titles of the chapters, section and subsection using decimal notation, as in the text, with corresponding page number against them, flushed to the right.

**3.7.1 List of Figures and Tables:** Two separate lists of Figure captions and Table titles along with their numbers and corresponding page numbers against them shall follow the Contents.

**3.8 Abbreviation Notation and Nomenclature:** A complete and comprehensive list of all abbreviations, notations and nomenclature including Greek alphabets with subscripts and superscripts shall be provided after the list of tables and figures. As far as possible, generally accepted symbols and notation should be used.

Auxiliary page from approval sheet to abbreviations shall be numbered using Roman numerals in lower case, while the text starting from the Introduction shall be in Hindu Arabic.

The first pages in the both the cases shall not bear a page number.

**3.9 A Declaration of Academic Honesty and Integrity:** A declaration of Academic honesty and integrity is required to be included along with every Project report after the approval sheet. The format of this declaration is given in Specimen ‘C' attached.

**Specimen 'A':**

Title Sheet (Title)

MAJOR PROJECT ON: < Title>

*Major Project Submitted in Partial Fulfillment of the Requirements for the degree of*

Master Of Computer Application
By
<Name Of the Student>
<university Roll Number>

Under the guidance of
<Name of The Guide(s)>




Techno Main, Salt Lake
EM 4/1 Salt lake City, Sector V
KOLKATA – 700091
<Year>

**Specimen `C' – Certificate for Approval**

(Maulana Abul Kalam Azad University of Technology)
Techno Main, Salt Lake
FACULTY OF MCA DEPARTMENT

## Certificate of Approval *

The foregoing Major project is hereby approved as a creditable study of Master of Computer Application (MCA) and presented in a manner satisfactory to warrant its acceptance as a pre-requisite to the degree for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or any statement made, opinion expressed or conclusion therein but approve this Major project only for the purpose for which it is submitted.

_____

_____

_____
Signature of the examiners

Final Examination for
Evaluation of the Project

- Only in case the Major project is approved

**Specimen `C' – Declaration**

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

_____

(Signature of the student)

_____

(Name of the student)

_____

(Roll No.)

Date: _____

**Specimen `D' – Certificate for Recommendation**

(Maulana Abul Kalam Azad University of Technology)
Techno Main, Salt Lake
FACULTY OF MCA DEPARTMENT

## Certificate of Recommendation

This is to certify that <name of the student> has completed his Major project (MCAN-482) work titled “Major project on: <Title>”, under the direct supervision and guidance of <Guide(s) Name>. We are satisfied with his work, which is being presented for the partial fulfillment of the degree of Master of Computer Application (MCA), Maulana Abul Kalam Azad University of Technology, Kolkata– 700064.

Name and signature of the HOD
Head of the Department
Department of MCA
TMSL

Name and signature of the Guide
Designation
Department of MCA
TMSL

## Appendix I

| Design Component | Points to Include | Description / Instructions |
|---|---|---|
| **Data Design** | 1. Data Requirements | Specify all data needed for the system, including input, output, and intermediate data. |
| | 2. Data Modelling | Include Entity-Relationship Diagrams (ERD), class diagrams, and data flow diagrams (DFD) where applicable. |
| | 3. Database Schema | Tables, fields, data types, primary/foreign keys, and relationships between tables. |
| | 4. Data Dictionary | Provide details of each data element: name, type, size, format, default value, and constraints. |
| | 5. Storage Considerations | Specify storage requirements, indexing, and data access methods. |
| **Architectural Design** | 1. System Architecture | Describe overall architecture. Include diagrams. |
| | 2. Component Design | List major modules/components and their responsibilities. |
| | 3. Interaction between objects | Sequence diagram |
| | 4. Technology Stack | Specify programming languages, frameworks, libraries, database, and tools used. |
| **Interface Design** | 1. User Interface (UI) Design | Show screen layouts, forms, menus, navigation flows, and graphical elements. |
| | 2. Input/Output Design | Specify how inputs are accepted (e.g., forms, file upload) and outputs displayed (e.g., reports, charts). |
| | 3. Interaction Design | Describe how users interact with the system, including user feedback and system prompts. |
| | 4. Consistency & Usability | Ensure design follows usability principles, consistency, and accessibility standards. |
| | 5. External Interfaces | Detail any interfaces with other systems, APIs, hardware, or third-party services. |

**Note**: When the MVC pattern is not employed, students should design the system using a **Data Flow Diagram (DFD)**. From the DFD, they must identify the **modules** of the system by analysing the major processes and their interactions. The **relationships between modules** should be clearly defined to ensure correct data flow and to minimise coupling. Additionally, a **Coordinator Module** must be designated to manage and coordinate interactions among the various modules, ensuring a smooth and logical operation of the system.

**APPENDIX II**

Students are required to include test cases for a specific module in their project report. The detailed test report should follow the sample format provided below. You must present your test cases in a clear and organised manner, ensuring that each test case includes the description, input data, expected output, actual output, pass/fail status, and any relevant remarks.

# Module Name: Login Module

**Objective of Testing:**
To validate all possible execution paths of the login module, ensuring correct authentication and appropriate error handling for valid and invalid inputs.

**Module Description:**

- **Inputs:** username, password
- **Outputs:** Success message or error message
- **Operations:** Input validation → Credential verification → Response generation

**Control Flow / Logical Steps:**

1. Start
2. Input username and password
3. If fields are empty → Display "Fields cannot be empty" → End
4. Else, validate credentials
   If valid → Display "Login Successful" → End
   If invalid → Display "Invalid Username/Password" → End

| Case ID | Description | Input Data | Expected Output | Actual Output | Pass /Fail | Remarks |
|---|---|---|---|---|---|---|
| TC1 | Both fields empty | username = ""password = "" | "Fields cannot be empty" | "Fields cannot be empty" | Pass | Validation works correctly |
| TC2 | Valid credentials | username = "user1"password = "pass123" | "Login Successful" | "Login Successful" | Pass | Authentication successful |
| TC3 | Invalid password | username = "user1"password = "wrongpass" | "Invalid Username/Password" | "Invalid Username/Password" | Pass | Error handled correctly |
| TC4 | Username not found | username = "unknown"password = "anypass" | "Invalid Username/Password" | "Invalid Username/Password" | Pass | Error handled correctly |
| TC5 | Special characters in input | username = "user!@#"password = "123$%" | "Invalid Username/Password" | "Invalid Username/Password" | Pass | Input sanitisation verified |

## APPENDIX III

### INTERFACE DESIGN FOR A SPECIFIC MODULE

**Assume: Interface one for module one**

| Interface Component | Type | Identifier / Name | Purpose / Functionality | Validation / Constraints | Input Format | Remarks |
|---|---|---|---|---|---|---|
| Label 1 (M1_I1_L1) | Label | lblUsername | Displays "Username" to indicate input field | N/A | N/A | Positioned above corresponding text box |
| Text Box 1 (M1_I1_T1) | TextBox | txtUsername | Allows user to input username | Cannot be empty; max 20 characters | `[A-Z][a-z][0-9]{1,20}` – any combination of letters and numbers, 1–20 chars | Positioned below Label 1 |
| Label 2 (M1_I1_L2) | Label | lblPassword | Displays "Password" to indicate input field | N/A | N/A | Positioned above corresponding text box |
| Text Box 2 (M1_I1_T2) | TextBox | txtPassword | Allows user to input password | Cannot be empty; max 20 characters; masked input | `[A-Z][a-z][0-9][!@#$%^&*()]{1,20}` – letters, numbers, and special characters, 1–20 chars | Positioned below Label 2 |
| Button (M1_I1_B1) | Button | btnLogin | Submits input data for processing | Enabled only when both text boxes are filled | N/A | Positioned below Text Box 2; triggers login validation |